

NUMBER SYSTEM

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NUMBER SYSTEM

REFERENCE BOOKS

- ❖ Digital Systems, Principle & Applications, Ronald J. Tocci, Prentice-Hall
- ❖ Digital Design, M. Morris Mano, Michael D. Ciletti, Pearson Education, Inc.
- ❖ Digital Circuits and Design, S. Salivahanan, S. Arivazhagan, Oxford University Press.
- ❖ Digital Electronics, G. K. Kharate, Oxford University Press.
- ❖ Digital Electronics, Bignell James, Logic and Systems, Cengage Learning
- ❖ Digital Logic and Computer Design, M. Morris Mano, Pearson Education, Inc.

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NUMBER SYSTEM

Many number systems are used in digital technology.

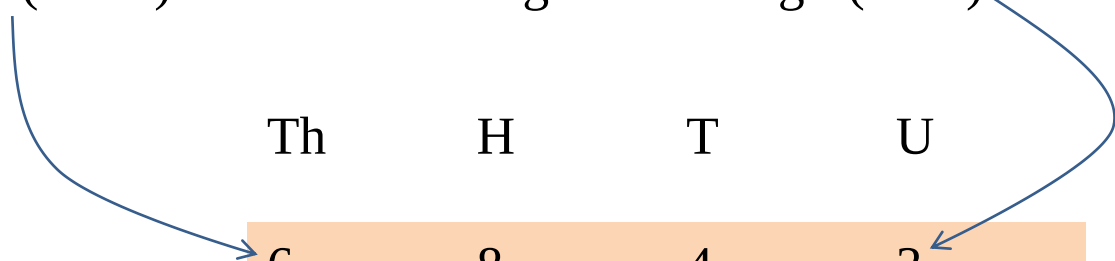
COMMON NUMBER SYSTEMS				
SYSTEM	BASE	SYMBOL	USED BY HUMANS	USED BY COMPUTERS
Decimal	10	0, 1, 2, 3... 9	Yes	No
Binary	2	0, 1	No	Yes
Octal	8	0, 1, 2, 3... 7	No	No
Hexa-decimal	16	0, 1, 2, 3... 9, A, B, ... F	No	No

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DECIMAL SYSTEM:

- The decimal number system is called Base-10 system because this system is composed of 10 numerals or symbols or digits.
- Ten digits used in decimal number system are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- The decimal number is composed of combinations of these digits.
- The decimal number is a positional-value system in which the value of a digit depends on its position. For example

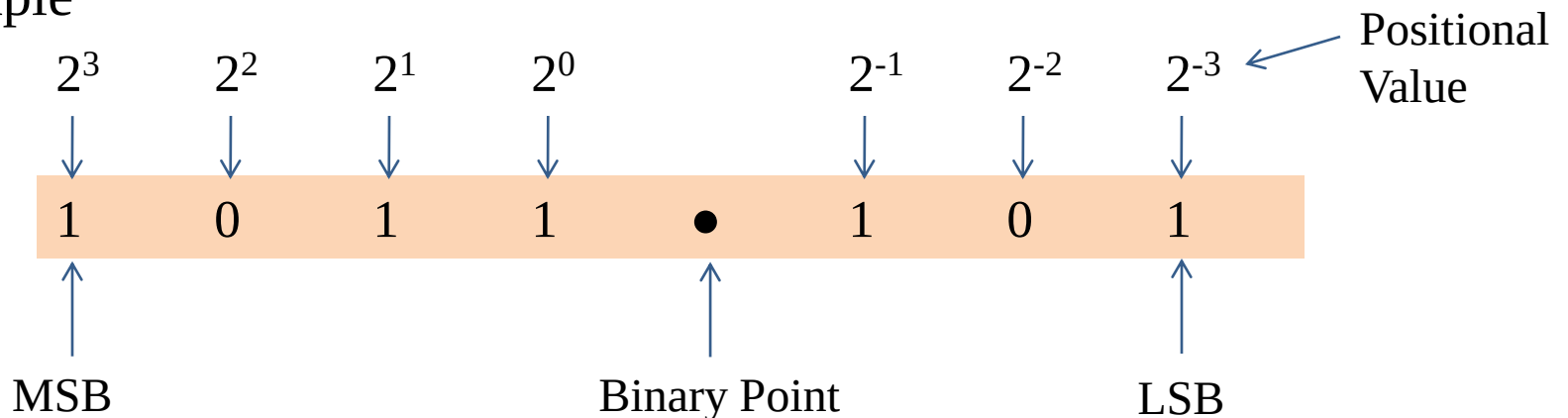
	Most Significant Digit (MSD)		Least Significant Digit (LSD)	
	Th	H	T	U
6843	6	8	4	3
Positional Values (Weights)	10^3 6000	10^2 800	10^1 40	10^0 3



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BINARY SYSTEM:

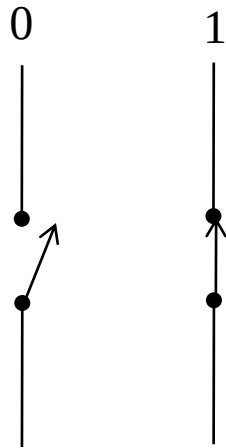
- In Binary number system, there are only two symbol or digits used.
- The base of the Binary number system is 2.
- The two digits used in this system are 0 and 1.
- These digits are called bits.
- In this system, Numbers are represented as string of bits.
- The Binary number system is also positional-value system, where each binary digit has its own value or weight expressed as a power of two. For example



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- Places to the left of the *binary point* (*counterpart of the decimal point*) are positive power of 2 and places to the right are negative powers of 2.

The most significant bit (MSB) is the leftmost bit (Largest weight). The Least significant bit (LSB) is the rightmost bit (smallest weight).



Switch has two states: open and closed. Open switch is represented by binary 0 and closed switch is represented by binary 1.

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OCTAL NUMBER SYSTEM:

- The representation of numbers corresponding to the large numbers in the decimal system are very long strings of zeros and ones. To overcome this disadvantage, a number system called Octal system is used to represent numbers.
- The octal number system has a base of 8, means that it has eight symbols: 0,1, 2, 3, 4, 5, 6, 7
- The octal number system is also a positional value system.
- Each octal digit has its own value or weight expressed as a power of 8.
- The places to the left of the octal point are the positive powers of 8 and places to the right are negative powers of 8. For example

Number: 4 3 1 • 5 1 0

Place Value: 8^2 8^1 8^0 8^{-1} 8^{-2} 8^{-3}

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HEXADECIMAL NUMBER SYSTEM:

- Hexadecimal number system uses 16 as a base.
- It uses sixteen symbols starting from zero.
- The representation of the digits from 0 to 9 are the same as decimal digits.
- To represent the decimal values from 10 to 15, Hexadecimal system uses the letters of alphabet A to F. For example

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

- The Hexadecimal number system is also a positional value system.
- Each Hexadecimal digit has its own value or weight expressed as a power of 16.
- The places to the left of the Hexadecimal point are the positive powers of 16 and places to the right are negative powers of 16. For example

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Number	A	B	6	•	C
Place Value	16^2	16^1	16^0		16^{-1}

THANK YOU

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